

L&T Machine M001

Maintenance and Safety Manual

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Sample document created for RAG testing. All data is fictional.

1. Purpose and Scope

This manual describes the safety rules, routine maintenance tasks, lubrication requirements, and troubleshooting procedures for the L&T Machine M001 hydraulic forming press. It is intended for maintenance technicians, shift supervisors, and safety officers. Operating instructions and technical specifications are covered separately in the M001 Technical Specification and Operation Guide (Document No. LT-M001-TS-04).

Maintenance work on the M001 is divided into three competence levels. **Level 1** tasks are daily and weekly checks that any trained operator may perform. **Level 2** tasks include filter changes, oil sampling, and minor adjustments, and must be performed by a technician who has completed the L&T Level 2 certification course. **Level 3** tasks include spindle and cylinder overhauls, calibration of the encoder and load cell, and changes to safety parameters; these may only be carried out by an authorised L&T service engineer.

2. General Safety Rules

The M001 can generate a pressing force of 800 kN and operates with hydraulic pressure of up to 210 bar. A hydraulic injection injury from a pinhole leak can be life-threatening even if the wound looks small. The following rules apply at all times:

- Never place any part of the body between the ram and the bed while the machine is energised or while the hydraulic system is pressurised.
- Never bypass, disable, or modify the light curtain, the two-hand control, or the door interlocks. Any suspected fault in a safety device must be reported immediately and the machine tagged out of service.
- Never search for hydraulic leaks with bare hands. Use a piece of cardboard or a leak-detection spray.
- Always use the mechanical ram support bar (the yellow safety prop) when working under the ram, even if the hydraulic system has been depressurised.
- Report all near-misses and incidents to the plant safety officer within 24 hours.

3. Personal Protective Equipment (PPE)

The minimum PPE for any person working within 2 metres of the machine is safety shoes with steel toe caps (rated to EN ISO 20345), safety glasses with side shields, and hearing protection when the machine is running, since the noise level is about 74 dB(A) and can rise during die-change work. When handling hydraulic oil, nitrile gloves must be worn. For work involving cutting, grinding, or handling of sharp dies, cut-resistant gloves rated at level C or above are required. Loose clothing, rings, and long unrestrained hair are prohibited around moving parts.

4. Lockout/Tagout (LOTO) Procedure

Before any Level 2 or Level 3 maintenance task, or before any task that requires opening a guard, the machine must be isolated using the following lockout/tagout procedure. Each person working on the machine must apply their own personal padlock to the isolation points.

Step	Action
1	Notify the shift supervisor and affected operators that the machine will be shut down for maintenance.
2	Complete the normal shutdown: finish the current cycle, select "Park" on the touchscreen, and wait for the message "Safe to power down".
3	Turn the main isolator to OFF and apply a personal padlock and a red danger tag.
4	Close the cooling water supply valve and lock it in the closed position if the cooling circuit will be opened.
5	Wait 10 minutes for the hydraulic accumulator to bleed down. Then open the manual bleed valve on the accumulator block and confirm that the accumulator pressure gauge reads 0 bar. The accumulator is pre-charged with nitrogen, so the gauge on the hydraulic side must be checked, not the nitrogen side.
6	Fit the yellow mechanical ram support bar before any person goes beneath the ram.
7	Attempt to start the machine (try-out) to verify that isolation is effective, then return all controls to the OFF position.

Only the person who applied a padlock may remove it. If the owner of a lock is absent, the lock may be removed only by the plant safety officer after a documented verification that the person is not working on the machine.

5. Preventive Maintenance Schedule

The schedule below is based on operating hours as shown on the SmartForm V3 hour meter. Intervals are cumulative: when an interval coincides with a shorter one (for example, at 2,000 hours, which is a multiple of 500 and 1,000), all tasks for the shorter interval must also be performed. Failure to follow the schedule may void the warranty.

Interval	Task	Level
Daily (start of shift)	Check oil level on sight glass; check for leaks; test emergency stop buttons and light curtain; wipe down the bed.	1
Weekly (every 50 hours)	Grease the four column guides; check the cooling water flow (minimum 25 L/min); clean the control cabinet air filter.	1
Every 500 hours	Check tightness of hydraulic fittings; inspect hoses for abrasion; verify accumulator pre-charge pressure (90 bar).	2
Every 1,000 hours	Replace the hydraulic return-line filter and the pressure-line filter; take an oil sample for laboratory analysis.	2
Every 2,000 hours	Replace the water-cooling circuit inline filter (50 micron); inspect the ram seals; check the foundation bolts torque (650 Nm).	3
Every 4,000 hours	Complete hydraulic oil change (320 litres of ISO VG 46); flush the reservoir; calibrate the encoder and load cell.	3
Every 20,000 hours	Major overhaul: replace main cylinder seals and ram guide bushes; inspect the frame for cracks.	3

6. Lubrication and Oil Management

Guide lubrication. The four chrome-plated columns are lubricated with NLGI Grade 2 lithium-complex grease. Each column guide receives 15 grams of grease per weekly service. Over-greasing can force grease into the die area and contaminate the work piece, so the quantity should be measured with a calibrated grease gun.

Hydraulic oil. Only ISO VG 46 hydraulic oil that meets the L&T approved list may be used. Mixing oils of different brands or types is prohibited, because incompatible additives can damage the seals and cause foaming. The oil must be changed every 4,000 operating hours or every 24 months, whichever occurs first. Used oil must be collected in labelled containers and disposed of through an authorised waste-oil contractor.

Oil cleanliness. The target cleanliness level for the hydraulic oil is ISO 4406 code 18/16/13. If the laboratory analysis taken at the 1,000-hour service shows a particle count above 20/18/15, the oil must be filtered through an offline filtration unit before the machine is returned to production. If the water content in the oil exceeds 0.1 percent, the oil must be replaced immediately, as water can cause pump cavitation and corrosion.

Filters. The M001 has two hydraulic filters: a return-line filter rated at 10 microns and a pressure-line filter rated at 5 microns. Each has a visual clogging indicator. If the indicator shows red at operating temperature, the filter must be replaced regardless of the number of hours since the last change. A cold start can temporarily trigger the indicator, so the check must always be made once the oil is above 40 degrees C.

7. Accumulator Maintenance

The bladder accumulator has a volume of 10 litres and a nitrogen pre-charge pressure of 90 bar. The pre-charge must be checked every 500 hours with the hydraulic side fully depressurised. If the pre-charge pressure has dropped below 80 bar, the accumulator must be recharged with dry nitrogen only. **Never use air or oxygen** to charge the accumulator, as this creates a risk of explosion. Recharging must be performed with a certified charging kit by a Level 2 or higher technician. If the bladder shows signs of damage, such as oil in the nitrogen side, the complete accumulator must be replaced by a Level 3 engineer.

8. Electrical Safety

The control cabinet contains voltages of up to 415 V and may only be opened by a qualified electrician after the main isolator has been locked out. After isolation, wait at least 5 minutes before touching any terminals in the servo drive section, because the drive capacitors can retain a dangerous charge. Use a multimeter to verify that the DC bus voltage is below 50 V before starting work. Insulation resistance tests must be performed annually and the results recorded in the maintenance log; the minimum acceptable value is 1 megohm.

9. Troubleshooting Guide

The SmartForm V3 control system displays warnings (prefix W), alarms (prefix A), and errors (prefix E). Warnings allow the machine to continue operating, alarms stop the machine at the end of the current cycle, and errors cause an immediate stop. The table below lists the most common codes.

Code	Description	Likely cause	Corrective action
W-12	Supply voltage out of range (+/- 10%).	Unstable grid; shared circuit breaker.	Check supply voltage; fit a stabiliser of at least 60 kVA.
A-31	Hydraulic oil temperature above 65 degrees C.	Insufficient cooling water; blocked heat exchanger.	Check water flow (min 25 L/min) and inlet temperature (max 30 degrees C); clean the inline filter.
A-35	Cooling water flow too low.	Closed valve; clogged 50 micron filter.	Open valve; clean or replace the inline filter.
E-47	Hydraulic pressure below 140 bar during pressing.	Clogged pressure-line filter; low oil level; worn pump; cold oil below 35 degrees C; accumulator pre-charge below 80 bar.	Check the oil level and filter indicator first; complete the warm-up cycle; check the accumulator pre-charge. If unresolved, call a Level 3 engineer to test the pump.
E-52	Encoder signal lost.	Damaged cable; dirty glass scale.	Inspect the cable; clean the scale with lint-free cloth and approved solvent. Re-calibration required (Level 3).
E-60	Light curtain beam interrupted or faulty.	Obstruction; misaligned unit; cable fault.	Clear the obstruction; realign the unit. Do not bypass the light curtain.
E-71	Ram position deviation too large.	Air in the hydraulic circuit; worn seals.	Bleed the circuit; inspect seals at next scheduled service.

Important note on E-47. Because E-47 involves the hydraulic circuit, the technician must complete the full lockout/tagout procedure in Section 4 before opening any hydraulic line, filter housing, or the accumulator block. In particular, the 10-minute accumulator bleed-down and the 0 bar confirmation on the hydraulic-side gauge must not be skipped, even if the machine has been switched off for a long time, because the accumulator can retain stored pressure indefinitely.

Repeated faults. If the same alarm or error occurs three times within one shift, the machine must be taken out of service and the fault reported to the maintenance supervisor. Operators must not reset an E-code more than twice without investigating the cause.

10. Calibration

The position encoder and the load cell must be calibrated every 4,000 operating hours, after any encoder replacement, and after any repair that involves removing the ram. Calibration is a Level 3 task and requires the L&T calibration kit, which includes a certified 800 kN reference load cell and a laser displacement gauge. After calibration the engineer will issue a calibration certificate that must be retained for the life of the machine. Changing the calibration data without authorisation is a breach of warranty conditions.

11. Recommended Spare Parts

The following parts are recommended to be held in stock at the customer site to reduce downtime.

Part	Part No.	Recommended Qty	Lead time
Return-line filter element (10 micron)	LT-F-1001	4	5 working days
Pressure-line filter element (5 micron)	LT-F-1002	4	5 working days
Light curtain unit	LT-S-2210	1	10 working days
Ram seal kit	LT-K-3305	1	15 working days
Accumulator bladder (10 L)	LT-A-4120	1	20 working days
Glass-scale encoder	LT-E-5008	1	30 working days

12. Records and Documentation

A maintenance log must be kept for each machine. Each entry must record the date, hour-meter reading, task performed, the name and competence level of the person who performed it, and any parts replaced. Records must be retained for at least 7 years, and must be presented to L&T on request for any warranty claim. Digital records stored in the SmartForm V3 event log are not sufficient on their own, because the log holds only the last 10,000 events.

13. Decommissioning and Disposal

At the end of its service life, the M001 must be decommissioned by an authorised contractor. All hydraulic oil (320 litres) must be drained and disposed of as hazardous waste. The accumulator must be fully discharged of nitrogen before removal. The steel frame, which is made of grade ST-52 cast steel, is fully recyclable. Electronic components, including the SmartForm V3 controller and the touchscreen, must be disposed of in accordance with local e-waste regulations.

14. Contact Information

For technical support, call the L&T Customer Support Centre, available Monday to Saturday from 8:00 AM to 8:00 PM IST. For emergencies involving injury or fire, always contact the plant emergency services first, and then notify L&T Customer Support within 24 hours. The guaranteed response time for critical breakdowns is 24 hours in India and 72 hours internationally.